JEE Chemistry

Detailed JEE Main + Advanced Syllabus

Exam Coverage	JEE Main + JEE Advanced aligned topic map
Format	Module-wise syllabus with detailed bullet-wise subtopics
Use Case	Foundation planning, course brochure, chapter scheduling and curriculum design

Structured Syllabus

Unit 1: Mole Concept, Atomic Structure and Periodicity

- Mole concept, stoichiometry, empirical and molecular formula, limiting reagent and concentration terms.
- Atomic structure, quantum numbers, electronic configuration, Bohr model and de Broglie concept.
- Periodic table, periodic trends, effective nuclear charge, atomic and ionic radii, ionization enthalpy and electron gain enthalpy.

Unit 2: Chemical Bonding and States of Matter

- Ionic, covalent and coordinate bonding, VSEPR theory, hybridization, resonance and molecular orbital basics.
- Gaseous and liquid states, gas laws, kinetic theory, real gases, critical constants and intermolecular forces.

Unit 3: Thermodynamics, Equilibrium and Redox

- First law of thermodynamics, enthalpy, entropy, Gibbs free energy and Hess law.
- Chemical and ionic equilibrium, Le Chatelier's principle, pH, buffer solutions, solubility product and hydrolysis.
- Redox reactions, equivalent concept and balancing in acidic/basic medium.

Unit 4: Solutions, Electrochemistry and Chemical Kinetics

- Types of solutions, concentration terms, Raoult's law, ideal and non-ideal solutions, colligative properties.
- Electrochemical cells, Nernst equation, conductance, Kohlrausch law, electrolysis and batteries.
- Rate law, order and molecularity, integrated rate equations, half-life and Arrhenius equation.

Unit 5: Inorganic Chemistry – Hydrogen, s/p/d/f Block and Coordination

- Hydrogen and its compounds; alkali and alkaline earth metals; periodic trends in s-block.
- Important compounds and reactions of p-block elements with JEE relevance including nitrogen, oxygen, halogens and noble gases.
- d- and f-block elements, oxidation states, coloured ions, lanthanoid contraction and qualitative trends.
- Coordination compounds: nomenclature, bonding theories, isomerism, stability and applications.

Unit 6: Organic Chemistry – General Principles and Hydrocarbons

- General organic chemistry: inductive effect, mesomeric effect, hyperconjugation, acidity/basicity and stability of intermediates.
- IUPAC nomenclature, structural, geometrical and optical isomerism; purification and qualitative analysis basics.
- Hydrocarbons: alkanes, alkenes, alkynes and aromatic hydrocarbons — preparation, reactions and mechanisms.

Unit 7: Organic Chemistry – Functional Groups

- Haloalkanes and haloarenes: substitution and elimination reactions, reactivity trends and named reactions.
- Alcohols, phenols and ethers; carbonyl compounds; carboxylic acids and derivatives with mechanism-based understanding.
- Amines, diazonium salts, biomolecules, polymers and chemistry in everyday life with application-oriented coverage.